Question:

A 17-year-old teenager is brought to the emergency department with severe bleeding from his right hand. He was involved in a gang fight about 30 minutes ago where he received a penetrating stab wound by a sharp knife in the region of the ‘anatomical snuffbox’. A vascular surgeon is called in for consultation. Damage to which artery is most likely responsible for his excessive bleeding?

Answer Choices:

A. Radial artery

B. Ulnar artery

C. Princeps pollicis artery

D. Brachial artery

Reasoning + Answer:

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Finding reasoning paths:

1. The anatomical snuffbox is located on the radial side of the wrist, between the extensor pollicis longus and brevis tendons.
2. The radial artery runs through the anatomical snuffbox.
3. Damage to the radial artery could result in significant bleeding due to its location and the high pressure of blood flow.
4. The ulnar artery is located on the medial side of the wrist and is less likely to be involved in bleeding from the anatomical snuffbox.

Reasoning Process:

1. **Anatomical Consideration:** The anatomical snuffbox is a triangular depression on the radial side of the wrist. It is bordered by the tendons of the extensor pollicis longus and brevis muscles. The radial artery passes through this area, making it vulnerable to injury in cases of trauma or penetration.

2. **Blood Supply and Pressure:** The radial artery is a major artery that supplies blood to the hand and forearm. It carries a significant amount of blood under high pressure, which can lead to substantial bleeding if damaged.

3. **Trauma and Bleeding:** Given that the teenager suffered a penetrating wound in the anatomical snuffbox, the most likely artery to be damaged is the radial artery, due to its anatomical location and the nature of the injury.

4. **Ulnar Artery Consideration:** The ulnar artery is located on the medial side of the wrist and is not part of the anatomical snuffbox. It is less likely to be involved in bleeding from a wound in the radial snuffbox area.

5. **Clinical Implications:** The high-pressure blood flow from the radial artery can lead to rapid and significant bleeding. This is a medical emergency that requires prompt attention, such as applying direct pressure to the wound and possibly involving surgical intervention.

Conclusion:

The bleeding is most likely caused by damage to the radial artery due to its anatomical location within the anatomical snuffbox and the high pressure of blood flow it carries.

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<answer>

Radial artery

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